

In programming, we often need to store information about a single real-world entity, such as a student, employee, car, or movie ticket booking. Using many separate variables for related information makes the program difficult to manage. To solve this problem, JavaScript provides **objects**, which allow us to group related data into a single unit.

Need for Objects

- To store related data together in one place.
- To reduce the use of multiple separate variables.
- To make programs easier to read, write, and maintain.
- To represent real-world entities with their attributes.
- To improve code organization and reusability.

An object is a collection(data structure) of related data stored as key–value pairs. Each key is called a property, and its corresponding value can be any data type such as a number, string, array, function, or even another object.

```
var customerName = "Ved";
var movieName = "Avengers";
var theatre = "PVR";
var seats = ["A1", "A2"];
var ticketPrice = 250;
var numberOfTickets = 2;

var totalAmount = ticketPrice * numberOfTickets;

console.log("Customer Name: " + customerName);
console.log("Movie Name: " + movieName);
console.log("Theatre: " + theatre);
console.log("Seats: " + seats);
console.log("Total Amount: ₹" + totalAmount);
```

Output

```
Customer Name: Ved
Movie Name: Avengers
Theatre: PVR
Seats: A1,A2
Total Amount: ₹500
```

```
var booking = {  
  customerName: "Ved",  
  movieName: "Avengers",  
  theatre: "PVR",  
  seats: ["A1", "A2"],  
  ticketPrice: 250,  
  numberOfTickets: 2  
};  
  
var totalAmount = booking.ticketPrice *  
booking.numberOfTickets;  
  
console.log("Customer Name: " + booking.customerName);  
console.log("Movie Name: " + booking.movieName);  
console.log("Theatre: " + booking.theatre);  
console.log("Seats: " + booking.seats);  
console.log("Total Amount: ₹" + totalAmount);
```

Output

Customer Name: Ved
Movie Name: Avengers
Theatre: PVR
Seats: A1,A2
Total Amount: ₹500

Modifying an Object

Modifying an object means **changing, adding, or deleting the properties or values of an existing object after it has been created.**

```
var student = {  
    name: "Ved",  
    age: 18  
};  
  
// Modify an existing property  
student.age = 19;  
  
// Add a new property  
student.city = "Vadodara";  
  
// Delete a property  
delete student.name;  
  
console.log(student);
```

Output

```
{  
  age: 19,  
  city: "Vadodara"  
}
```

```
var student = {  
    name: "Ved",  
    marks: [65, 70, 75, 80]  
};  
  
console.log("Original Marks: " + student.marks);  
  
var i = 0;  
while (i < student.marks.length) {  
    student.marks[i] = student.marks[i] + 5;    // Modify each mark  
    i++;  
}  
  
console.log("Updated Marks: " + student.marks);
```

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Output

Original Marks: 65,70,75,80

Updated Marks: 70,75,80,85

```
var product = {  
    name: "Laptop",  
    prices: [45000, 50000, 55000]  
};  
  
console.log("Original Prices: " + product.prices);  
  
var i = 0;  
  
while (i < product.prices.length) {  
    product.prices[i] = product.prices[i] + 2000;  
    i++;  
}  
  
console.log("Updated Prices: " + product.prices);
```

Output

Original Prices: 45000,50000,55000

Updated Prices: 47000,52000,57000